



HYPOTHETICAL TRADE ANALYZER

- By SeniHers Group 1

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DESCRIPTION

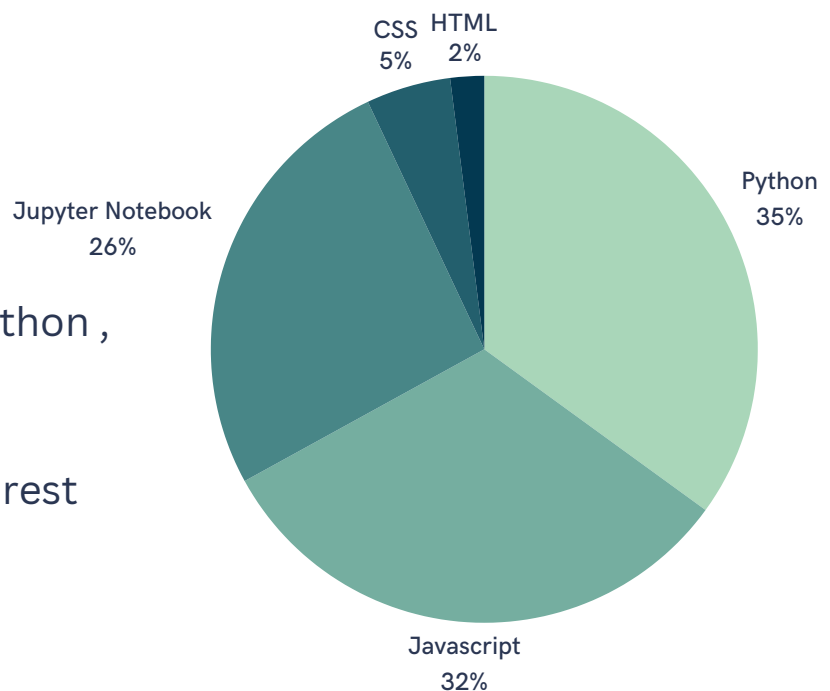
The **Hypothetical Trade Analyzer** , is a sophisticated web-based platform designed to revolutionize the way investors manage their trading portfolios.

Offering a seamless user experience, the platform allows registered users to access their accounts, view transaction history, and analyze trade performance with precision.

By integrating cutting-edge algorithms, the analyzer computes real-time profit and loss (P&L) figures, empowering users to make informed decisions and optimize their investment strategies.

Tech Stack

- **Programming Languages** : Python , Bootstrap , Javascript .
- **Data Storage** : SQLite
- **Backend Framework** : Django rest
- **Frontend Framework** : React
- **API** : Yahoo Finance



REQUIREMENTS

- **User Authentication and Registration:** The platform offers secure user authentication mechanisms, allowing investors to create accounts and access personalized features.
- **Real-Time P&L Calculation:** With real-time **P&L calculation capabilities**, users can instantly assess the financial implications of their trading decisions. By understanding the potential profitability of each trade, investors can make timely adjustments to their portfolios and capitalize on market opportunities.
- **Transaction History Management:** Users can effortlessly track their trading activities by maintaining a comprehensive transaction history within the platform. This feature enables them to review past trades and assess overall portfolio performance.
- An algorithm to **retrieve** a user's **transaction history** from the database and show it on the history page.
- Scripts to analyze the **performance** of the funds, the **profits or losses** **they** produced, and present them as **charts**.



ALGORITHM

For Risk Calculation

The risk calculation algorithm for the Hypothetical Trade Analyzer involves assessing the impact of adding a new stock to a user's portfolio on the overall portfolio risk and variance.

This algorithm uses historical price data retrieved through the Yahoo Finance API and quantities of stocks bought by the user, it calculates the portfolio risk associated with adding a new stock to the existing portfolio.

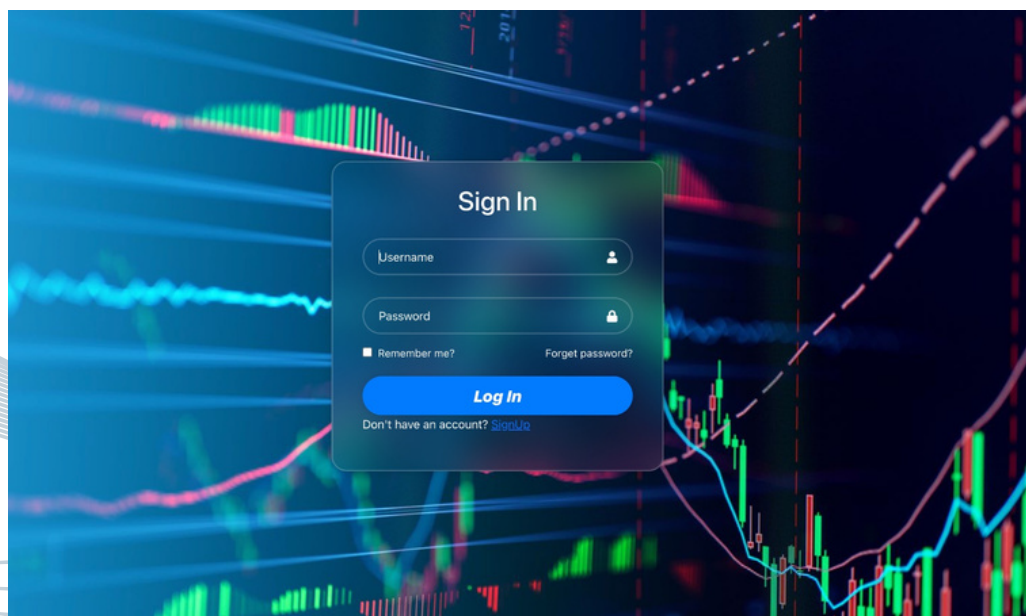
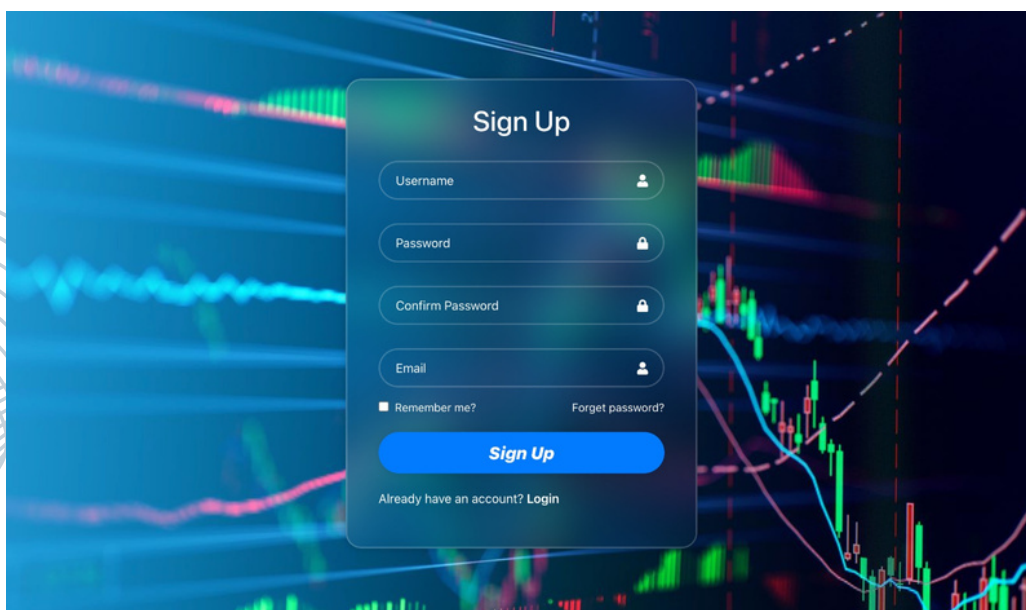


ALGORITHM STEPS:

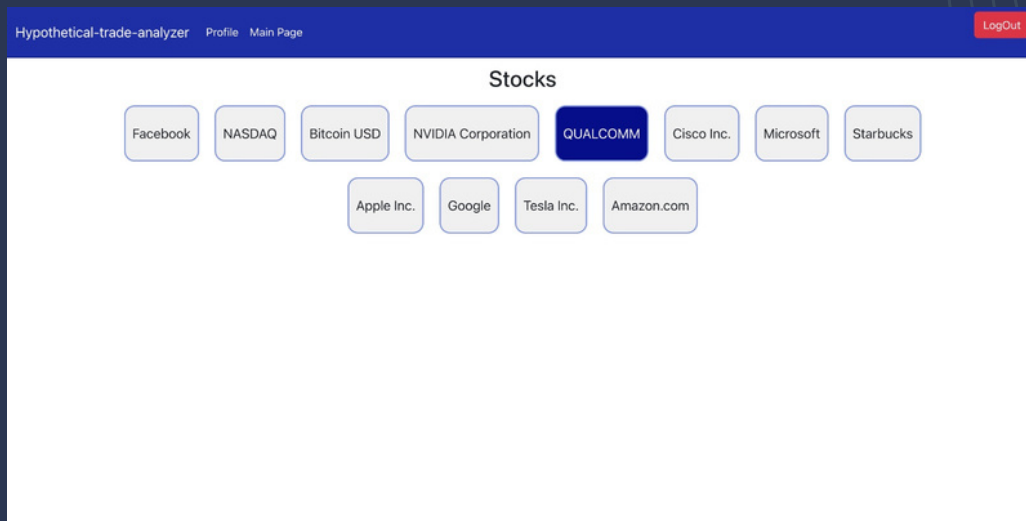
1. Retrieve User's Portfolio:
 - a. Extract the list of stocks bought by the user from the database.
 - b. Obtain the quantities of each stock bought by the user.
2. Add New Stock:
 - a. Check if the new stock to be analyzed is already present in the user's portfolio.
 - b. If not present, add the new stock to the list of stocks bought by the user.
 - c. Update the quantity of the new stock accordingly.
3. Retrieve Historical Price Data:
 - a. Utilize the Yahoo Finance API to fetch the previous 10 prices for each stock in the portfolio, including the newly added stock.
4. Compute Percentage Change in Prices:
 - a. Calculate the percentage change in price for each entry in the historical price data of all stocks.
5. Compute Covariance Matrix:
 - a. Formulate a covariance matrix of the percentage changes in prices for all stocks.
 - b. The covariance matrix captures the relationships and dependencies between the stocks' price movements.
6. Determine Stock Weights:
 - a. Assign weights to each stock based on the ratio of quantities held by the user.
 - b. Ensure that the sum of weights equals 1, representing the proportionate allocation of capital across the portfolio.
7. Calculate Portfolio Risk:
 - a. $\text{P\&L covariance} = \text{np.dot(weights, np.dot(covariance_matrix, weights.T))}$
 - b. Portfolio risk is the square root of the variance, representing the standard deviation of portfolio returns.

UI Snippets

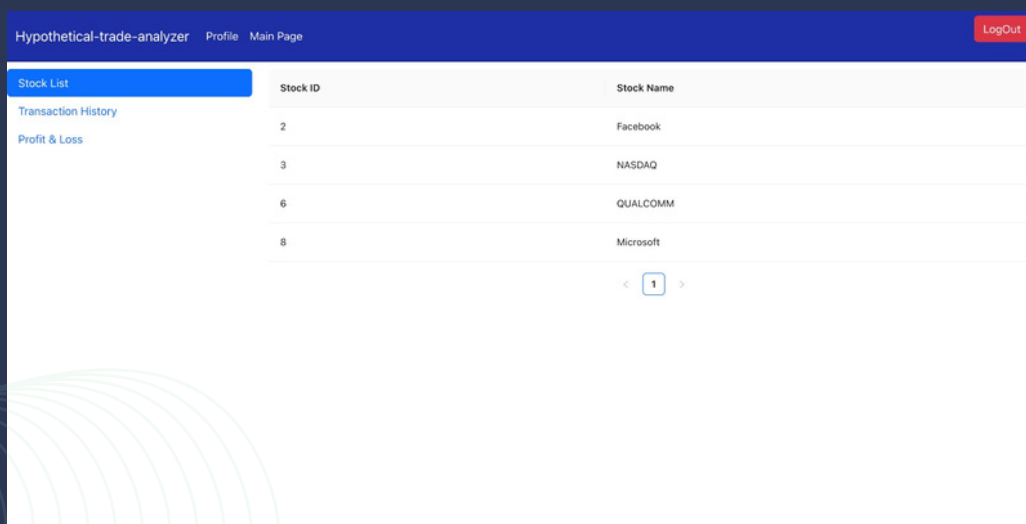
- **User Authentication and Registration:** The platform offers secure user authentication mechanisms, allowing investors to create accounts and access personalized features.



- **Profile page:** It offers users a curated selection of stocks available for investment, providing valuable insights into potential opportunities. Additionally, users can hypothetically assess the profit and loss (P&L) of each stock, enabling them to gauge potential returns and make informed investment decisions.



- **Stock List:** The Stock List page provides investors with a comprehensive overview of their current holdings, displaying all stocks in which they have invested.



- **Transaction History page:** It presents users with a detailed chronological record of all their stock transactions, organized by date. Users can view essential details such as market value, price, quantity, and other relevant information for each transaction.

Hypothetical-trade-analyzer Profile Main Page LogOut

Stock List	ID	Date	Quantity	Price	Market Value	Stock ID
Transaction History	1	2024-03-19	3	45	135	2
Profit & Loss	2	2024-03-19	3	45	135	2
	3	2024-03-19	3	45	135	2
	4	2024-03-19	2	45	90	2
	5	2024-03-19	6	45	270	2
	6	2024-03-21	3	45	135	
	7	2024-03-22	33	171	5655	
	8	2024-03-22	3	171	514	
	9	2024-03-22	33	171	5655	
	10	2024-03-22	3	171	514	

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Hypothetical-trade-analyzer Profile Main Page LogOut

Stock List	ID	Date	Quantity	Price	Market Value	Stock ID
Transaction History	11	2024-03-22	3	171	514	3
Profit & Loss	12	2024-03-22	3	171	514	3
	13	2024-03-22	5	171	856	3
	14	2024-03-22	10	170	1700	6
	15	2024-03-22	10	427	4277	8

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- **Profit and Loss:** The Profit and Loss page provides users with a detailed breakdown of their profit or loss for each stock purchase, displayed by date. It also offers a summary of the total profit or loss across all investments, offering insights into overall trading performance.

Hypothetical-trade-analyzer Profile Main Page LogOut

Stock List	Stock ID	Profit and loss	Date
Transaction History	2	-7538	2024-03-21
Profit & Loss	4	257880	2024-03-22
	3	-204449	2024-03-22
	6	-1704	2024-03-22
	8	-4280	2024-03-22

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Total Profit and loss

48488

localhost:3000/profile#

- **Risk Calculation:** This page enables users to select a specific quantity of stock and assess the associated risk, profile variance, and potential profit or loss (P&L). Users can compare previous results with new calculations to analyze changes in risk exposure and investment outcomes.

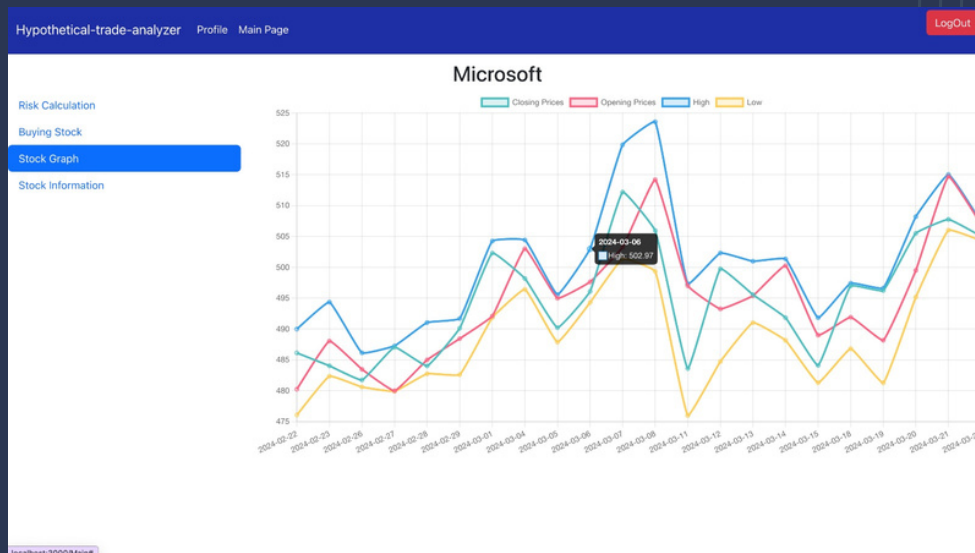
The screenshot shows the 'Risk Calculation' page for Microsoft. The page has a blue header with 'Hypothetical-trade-analyzer', 'Profile', 'Main Page', and a 'Logout' button. A sidebar on the left contains 'Risk Calculation' (highlighted), 'Buying Stock', 'Stock Graph', and 'Stock Information'. The main content area is titled 'Microsoft' and 'Quantity'. It features a text input field with '10' and a 'Submit' button. Below this is a table comparing 'Old' and 'New' values for three attributes: Risk, Portfolio Variance, and PnL. The 'Old' values are 2.9, 9.38, and 0 respectively, while the 'New' values are 3.08, 9.47, and 0. A pagination control at the bottom shows '< 1 >'.

Attribute	Old	New
Risk	2.9	3.08
Portfolio Variance	9.38	9.47
PnL	0	0

- **Buying Stock:** The Buying Stock page simplifies the stock purchasing process, allowing users to select the desired quantity and execute their buy orders seamlessly.

The screenshot shows the 'Buying Stock' page for Microsoft. The page has a blue header with 'Hypothetical-trade-analyzer', 'Profile', 'Main Page', and a 'Logout' button. A sidebar on the left contains 'Risk Calculation', 'Buying Stock' (highlighted), 'Stock Graph', and 'Stock Information'. The main content area is titled 'Microsoft' and 'Buying Stock'. It features a text input field with '10' and a 'Submit' button. Below this is a green checkmark icon and the text 'Successfully Calculated !'. A message below the checkmark states 'The Transaction has been successfully Updated.' At the bottom, there are two buttons: 'Done' and 'Calculate Again'.

- **Stock Graph:** The Stock Graph page offers users an insightful visualization of a particular company's stock performance. Through interactive graphs, users can track key metrics such as opening and closing prices over time.



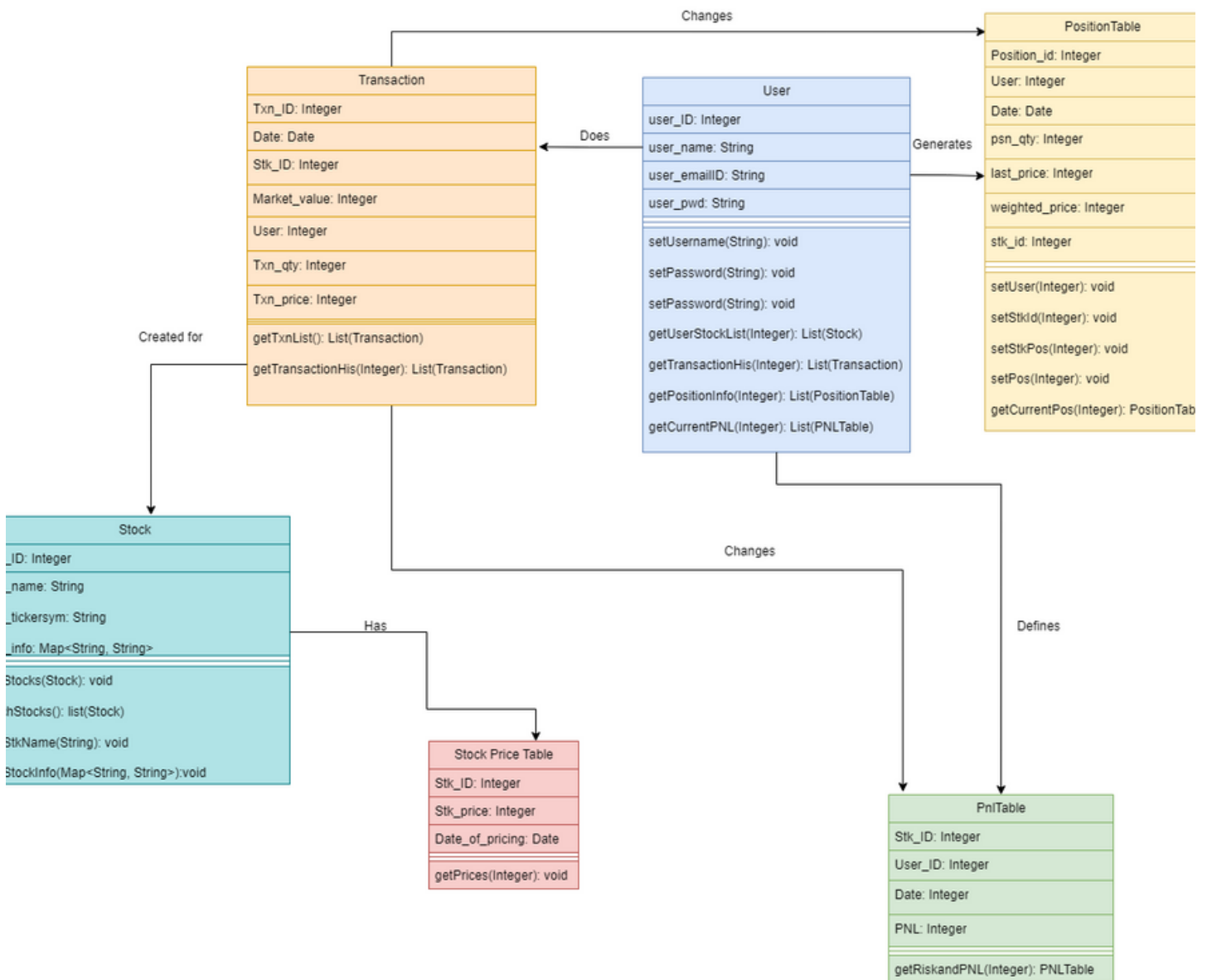
- **Stock Information:** The Stock Information page provides users with comprehensive details about a specific stock, including key metrics such as current price, trading volume, market capitalization, dividend yield, and earnings per share.

Microsoft		
Stock Attribute Name	Value	
Company Sector	Computational Services	
Price Earnings Ratio	33.864024	
Company Beta	1.204	
PE ratio	33.867005	
Dividend Yield	0.0039	
Market Cap	1217568760300	
Volume	470258	
Average Volume	1848161	
Previous Close	507.76	
Open	507	

Microsoft		
Stock Attribute Name	Value	
High	507.96	
Low	504.84	
52 Week High	519.51	
52 Week Low	480.8	
50 Day Moving Average	492.3724	
200 Day Moving Average	485.1023	
Price to Sales Ratio	9.748446	
Price to Book Ratio	8.4488439	
Currency	USD	

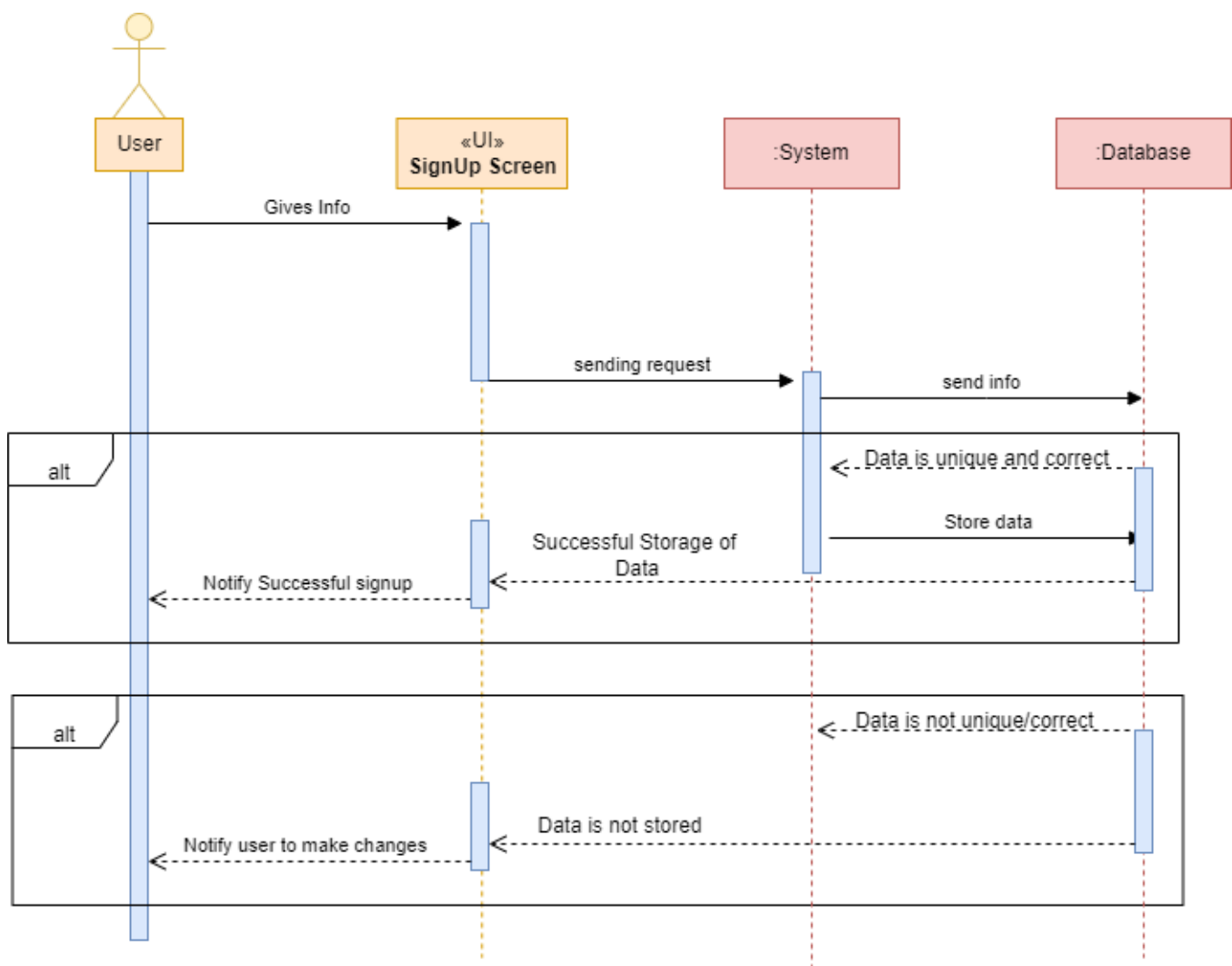
DESIGN AND ARCHITECTURE

USER CLASS DIAGRAM

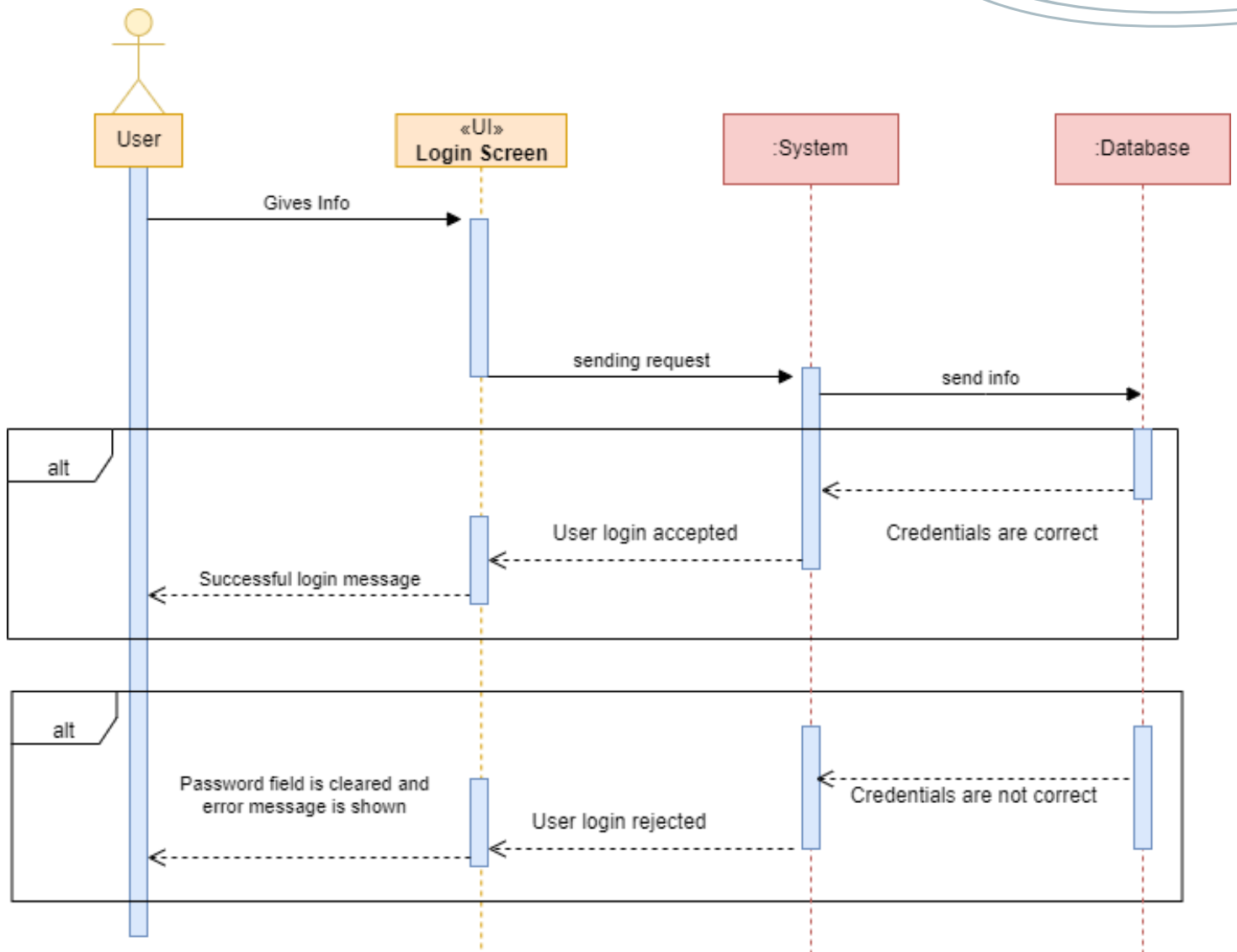


SEQUENCE DIAGRAM

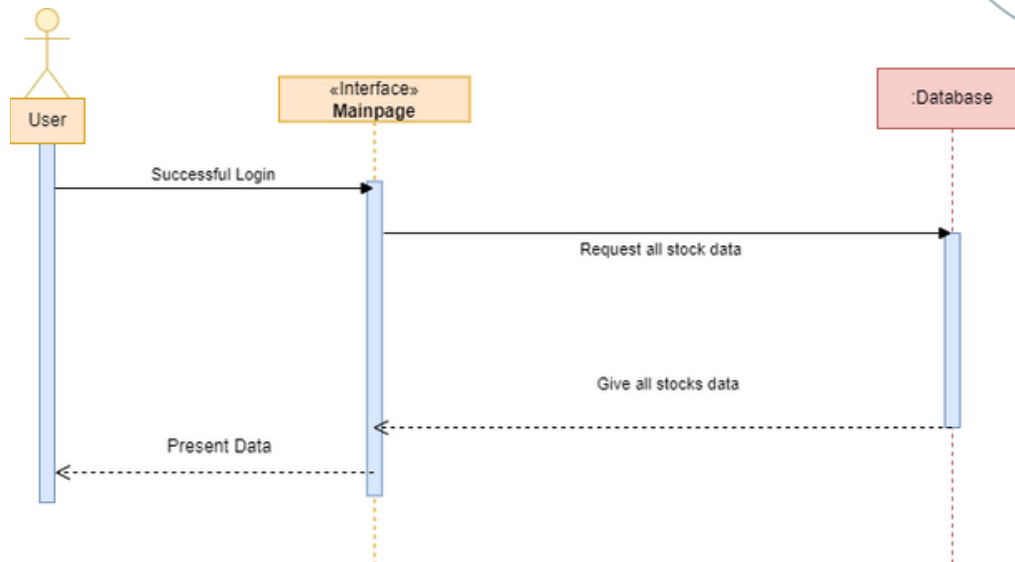
1. User Requests for SignUp



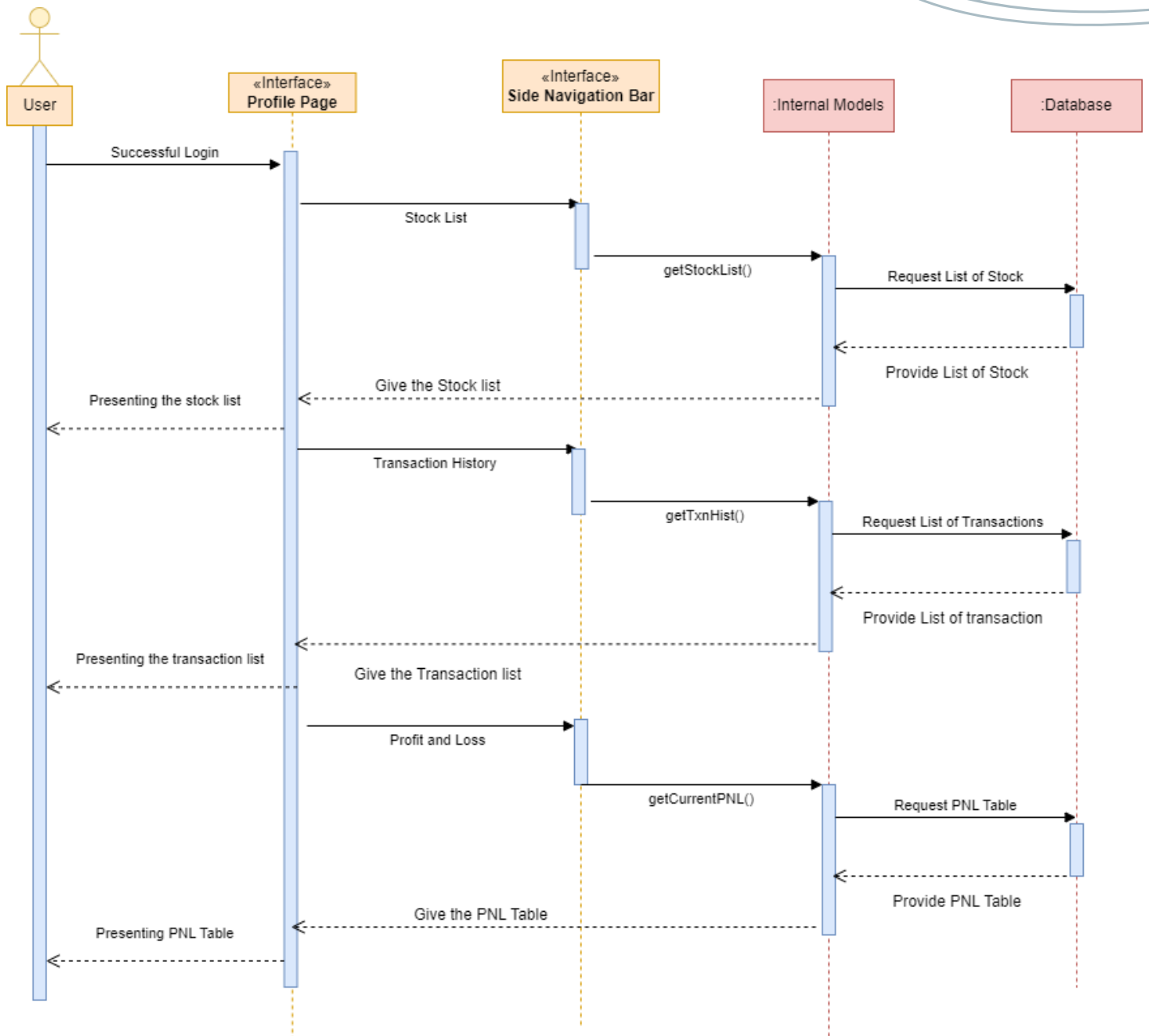
2. User Requests for Login



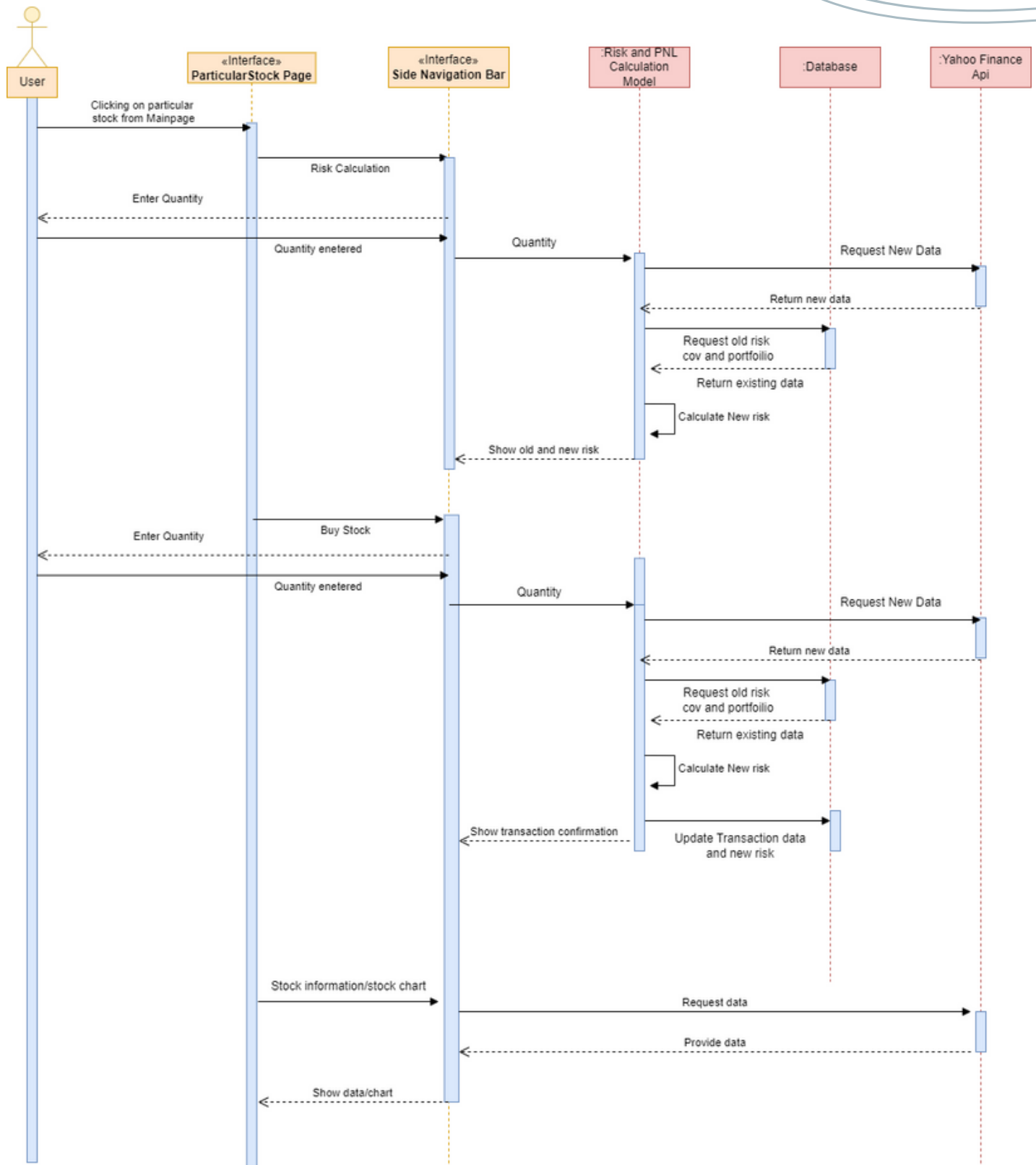
3. Mainpage



4. Profile Page

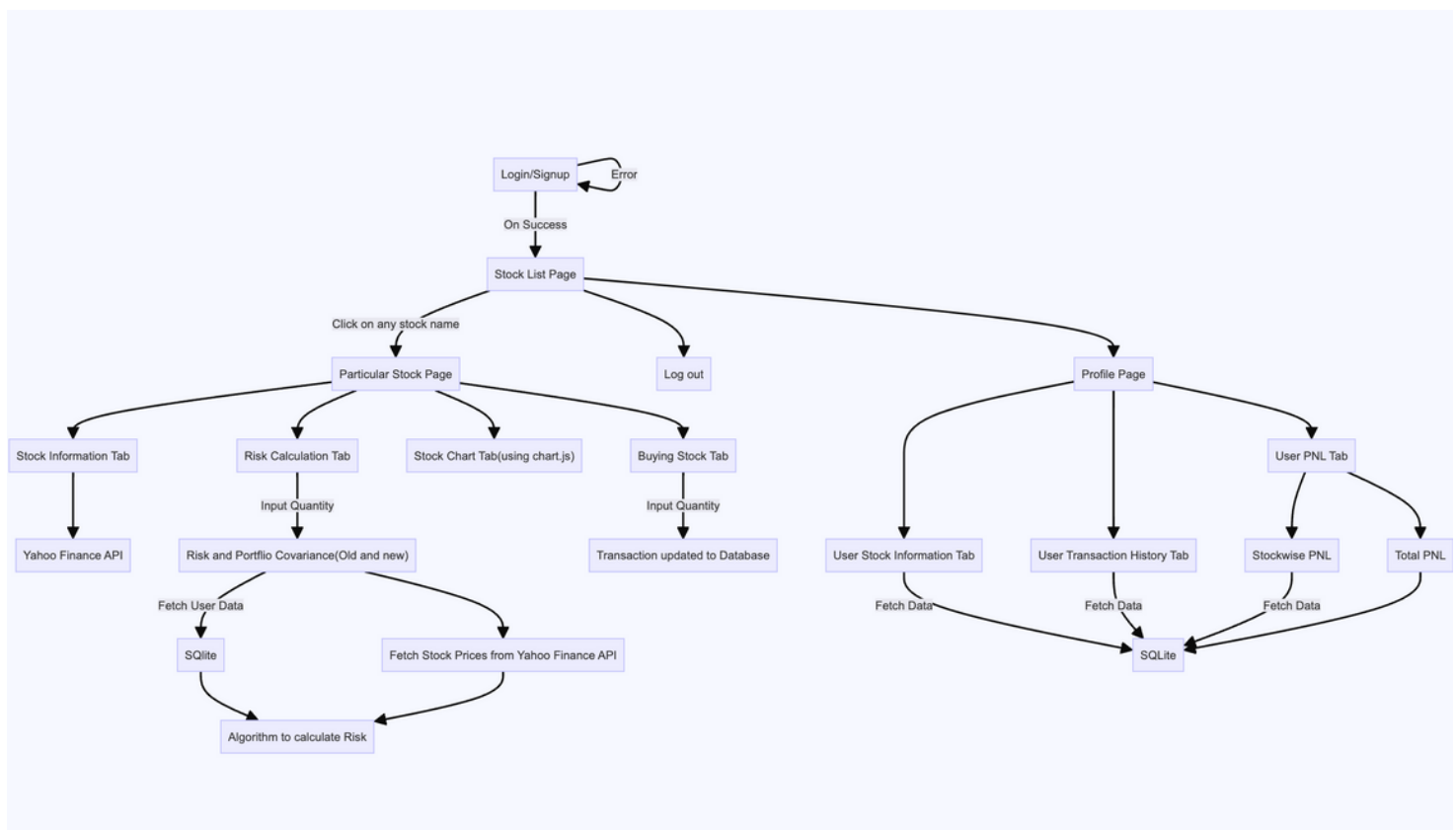


5. Particular Stock Page



SYSTEM DESIGN

The following flowchart illustrates the user journey through our website. This flowchart outlines the steps users take from logging in or signing up to accessing various features such as the stock list page, profile page, and specific stock information. It also details backend processes such as data fetching from SQLite and Yahoo Finance API, risk calculation, transaction updates, and PNL (Profit and Loss) calculation.



CODE QUALITY

CODE MODULARITY

The backend has been compartmentalized into distinct modules, each tasked with handling specific functionalities of the program. Each module is self-contained, encompassing all components required to execute a singular aspect of the desired functionality.

- **Settings:** The settings.py file encapsulates all configuration settings for the entire project, such as database settings, middleware configuration, and third-party app integration. By centralizing configuration, it promotes reusability and consistency across the project.
- **URLs Configuration:** The urls.py file defines URL patterns for the entire project. Mapping URLs to views or other URL configurations allows for a modular and flexible routing system.
- **Models:** The models.py file within each application defines data models specific to that application. Models encapsulate data structure and behaviour, promoting encapsulation and reusability.
- **Views:** Views, defined in the views.py file, encapsulate request handling logic. Views are typically designed to be modular and reusable, facilitating code organization and maintainability.

- **Admin Interface:** The `admin.py` file allows models to be registered with Django's admin interface. This promotes modularity by separating administration-related code from the rest of the application logic.
- **Serializers:** Serializers convert complex data types into formats suitable for transmission over the web, promoting modularity by separating data conversion from view logic. `Serializers.py` file is separately created to handle model serialization and support custom serialization, enhancing code reuse and maintaining separation of concerns.
- **Utils:** Utility modules, or "utils," contain reusable functions and classes that perform similar tasks across the project. `Util` file promotes modularity by centralizing standard functionality, enhancing code readability, and improving testability through isolation and abstraction of similar tasks.

CLEAN CODE

Clean coding practices are meticulously applied throughout the project, ensuring readability, maintainability, and efficiency.

- The codebase strictly follows the guidelines outlined in PEP 8, promoting consistency and readability across all modules.

- Descriptive names are consistently employed for variables, functions, classes, and modules, enhancing code comprehension.
- Functions and methods are kept succinct, focusing on single responsibilities. Clear comments elucidate complex logic when needed.
- Effective use of whitespace and consistent indentation improve code readability, while adherence to formatting standards is ensured using tools like Black.
- Duplicated code is minimized through the extraction of common functionality into reusable components, promoting maintainability and reducing the risk of errors.
- Exception handling is implemented gracefully, with informative error messages and appropriate abstraction levels, ensuring robustness and resilience.
- Python's rich standard library is leveraged wherever possible, reducing complexity and promoting code efficiency.
- The codebase prioritizes simplicity and clarity over cleverness, avoiding unnecessary complexity that may hinder understanding and maintenance.
- Changes to the codebase are tracked using a version control system like Git, with frequent commits and meaningful commit messages ensuring traceability and collaboration.

FUTURE SCOPE

- Adding Selling stock functionality.
- Exploring More Risk Algorithms .
- Mobile Trading App.
- Enhanced User Experience.
- Expanded Market Coverage: Extend the range of markets covered by the platform to include international stock exchanges, commodities markets, cryptocurrencie .
- Educational Resources:Expand the educational resources section to provide users with access to tutorials, articles, and online courses covering various aspects of trading, investing strategies, technical analysis, and financial literacy.



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